

it is free, it can decrease the TCO (total cost of ownership) of the phone. Given that Android works from within an open community initially formed by OHA (Open Handset Alliance) and now by Google, the abundance of freely available software for handheld devices, or rather Android-based devices, should increase once it is launched in the market.

Architecture overview

Android has a layered architecture with the total abstraction of the kernel/drivers from the application programs. The stack mainly consists of four layers. The bottom-most layer consists of the Linux kernel and device drivers. The kernel communicates with the libraries layer. The second layer is the libraries layer, which is made up of the various libraries of Android like standard graphics libraries (OpenGL), SSL, SQLite (a database library), etc.

Apart from the libraries, this layer also contains the most important part of the stack, commonly known as the Android Runtime or the DVM (Dalvik Virtual Machine). Just like the JVM, DVM abstracts applications programs from the kernel. This results in platform independency, so an application once written for Android can be run on any handheld device running Android, whatever hardware it may be running on.

The third layer is the application framework layer, which has pre-written modules for the phone, like an activity manager, window manager, telephony manager, notification manager, location manager, etc. If we draw an analogy to the Java world, Android runtime can be extended to the JRE library (Java Runtime Environment). The application framework is responsible for exposing the API for application development.

On top of all these three layers is the final layer—the applications themselves. For example, a phone dialler is an application that uses the telephony management API from the application framework layer. This, in turn, will use a core library of the stack and, in turn, the serial device driver from the Linux kernel to make a call using serial commands to communicate with the GSM-based SIM card.

An architecture diagram of Android is available on Google's site for Android at code.google.com/android/what-is-android.html, which has been reproduced here in Figure 1 for easy reference.

Pre-requisites

Instead of going into the depths of each library, we will focus on developing an application on the Android platform. We will focus on running the application on a simulator.

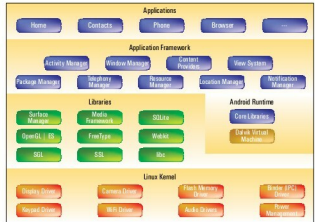


Figure 1: The Android architecture

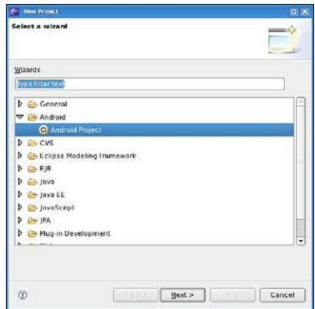


Figure 2: From the New Project wizard, select Android Project

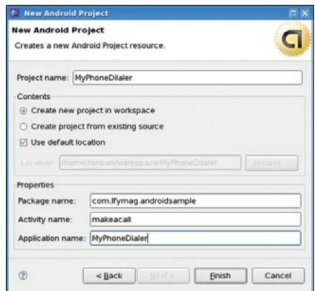


Figure 3: Define project name in the New Android Project wizard